

PM6750
Multiparameter Monitor Module(Bluetooth)
Tech manual

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www.spo2.cn

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1. introduction

PM6750: six-parameter bluetooth moduel is the new designed one by Shanghai Berry which is including Spo2,pulse rate,ECG, NIBP and resp functions.It can be connected by serial RS232, bluetooth and usb.

2. Specifications

2.1 Physical Dimensions

Length:12cm

Width:12cm

Height:2cm

Weight:80g

2.2 *Environment*

Temperature

Operation temperature	5~40°C
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Transportation and storage temperature	-20~55°C
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Humidity

Operation Humidity	≤85%
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Transportation and storage humidity	≤93%
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Height above sea level

Operation height above sea level: -500~4,600 meters

Transportation and storage height above sea level: -500~13,100 meters

✧ 2.3 *Specification*

ECG

✧ Full lead: I, II, III, aVL, aVR, aVF and V1, V2, V3, V4, V5, V6

✧ Gain:2.5mm/mV, 5mm/mV, 10mm/mV, 20mm/mV

✧ Scanning speed: 12.5mm/s, 25mm/s and 50mm/s

✧ Band width:	0.05Hz-100Hz(3dB)	Diagnostic mode
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	1-40Hz	Monitor mode
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	0.1-25Hz	Operation mode
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✧ Heart rate:

Scope: 0 times/minute to 250 times/minutes

Accuracy: ± 2 time/minute

SpO2

✧ Measurement scope: 0%-100%

✧ Accuracy: $\pm 2\%$ in the range of 90%~100%

$\pm 3\%$ in the range of 70%-89%

NIBP

✧ Measurement method: Electric oscillator

✧ Cuff inflation time: less than 15 seconds

✧ Initial inflation pressure: Dynamic state 24kPa (180 mmHg)

✧ Measurement range:

Systolic blood pressure: 4kPa -34kPa (30mmHg -255mmHg)

Diastolic pressure: 2kPa -29.3kPa (15mmHg -220mmHg)

Mean pressure: 2.7kPa~31.3kPa (20mmHg~235mmHg)

✧ Measurement accuracy:

Static pressure: ± 3 mmHg

Resp

✧ Measurement methods: Chest impedance method or concentration of carbon dioxide at end of respiration

✧ Range of chest impedance method: 0 time/minute -100 minutes/minute

✧ Accuracy of chest impedance method: ± 2 time/minute

Temp

✧ Single/double path

✧ Type: Vascular/surface temperature

✧ Scope: 20°C ~50°C

✧ Accuracy: $\pm 0.2^\circ\text{C}$

Communication settings

RS232 or TTL Level ,Bluetooth ,USB

Baud rate:115200

- non parity
- 8 data bits
- 1 stop bit

RS232 Command

Sign bit	Data length	Data	Sum
0x55 0xAA	N	A1, A2, ..., An	
SUM			

Means :

- 1、Data sign bit: 0x55 0xAA;
- 2、Dage length: $N = n + 2$, from 3~255
- 3、Data: include n bite;
- 4、SUM = $\sim (N + A1 + A2 + \dots + An)$

Communication protocol

1、PC→PM6750 (control format)

Data command	A1 (the command head)	A2 (the command)
ECG control	0x01	command
NIBP control	0x02	command
Spo2 control	0x03	command
Temperatur control	0x04	command
ECG waves Set	0x07	command
ECG mode set	0x08	command
NIBP mode set	0x09	command
NIBP target pressure set	0x0A	command

Static pressure Test	0x0B	command
Static pressure calibration set	0x0C	command
Temp1 calibration command	0x0D	command
Temp2 calibration command	0x0E	command
Respiration Gain set	0x0F	command
NIBP leak test	0x10	command
The version of software read	0xFC	command
The version of hardware read	0xFD	command
The Spo2 waves out set	0xFE	command
The Respiration waves out set	0xFF	command

detailed description:

ECG control (0x01) : control the ECG waves, data;

A2 (command) can use :

0x00 No data output

0x01 ECG data can output

NIBP control (0x02) : control the NIBP function;

A2 (command) can use :

0x00 stop the NIBP test

0x01 start the NIBP test

Spo2 control (0x03) : control the Spo2 function;

A2 (command) can use :

0x00	no Spo2 data output
0x01	Spo2 data can output

Temperature control (0x04) : control the temperature function;

A2 (command) can use:

0x00	no Temp data output
0x01	The Temp data can output

ECG gain set command (0x07) : set the Gain of ECG waves

A2 (command) can use :

0x01	x0.25 Gain
0x02	x0.5 Gain
0x03	x1 Gain
0x04	x2 Gain

Control the ECG mode (0x08) : set the ECG mode

A2 (command) can use:

0x01	operation mode, 1 ~ 25Hz (3dB)
0x02	Monitor mode, 0.5 ~ 75Hz (3dB)
0x03	diagnosis Mode, 0.05 ~ 100Hz (3dB)

NIBP mode control (0x09) : set the NIBP test mode

A2 can use :

0x01 adult mode (default mode)

0x02 Chile mode

0x03 neonate mode

NIBP target pressure set (0x0A) : set the NIBP target pressure value

A2 can use :

Test mode	Target pressure set (mmHg)
Adult mode	40~300 mmHg (default 150)
Child mode	40~210 mmHg (default 100)
neonate mode	40~140 mmHg (default 70)

attention:

1)、the pressure = Set value / 2, mmHg

2)、Before the start test ,use this command,or else the system use the default pressure value

3)、The range of pressre set value must in the mode such as Child and neonate mode

Static pressure Test command (0x0B) : not use for people ! close or open the air valve

A2 can use:

0x00 stop the static pressure test

0x01 start the static pressure test,close the air valve

Static pressure calibration set (0x0C) : just for calibration for manufacture !

A2 can use : -50 ~ +50, means -50mmHg ~ +50mmHg

Temp1 calibration command (0x0D) : just for manufacture calibration !

A2 can use : -20 ~ +20, means -2.0°C ~ +2.0°C ,

Temp1 calibration command (0x0E) : just for manufacture calibration !

A2 can use : -20 ~ +20, means -2.0°C ~ +2.0°C ,

Respiration waves Gain set (0x0F) : set the Gain of respiration waves

A2 can use :

0x01 x0.25Gain

0x02 x0.5Gain

0x03 x1Gain

0x04 x2 增益

NIBP leak test command (0x10) : Just for test the system !

A2 can use:

0x00 stop the leak test

0x01 pressure valve start the leak test : mmHg

The version of software read (0xFC) : for get the version of software

A2 not used.

The version of hardware read (0xFD) : for get the version of hardware

A2 not used

The Spo2 waves out set (0xFE) : control the Spo2 waves output

A2 can use:

0x00 no Spo2 waves data output

0x01 The Spo2 waves data can output

The Respiration waves out set (0xFF) : control the Respiration waves output

A2 can use:

0x00 no Respiration waves output

0x01 the Respiration waves data can output

PC to module command Examples

Start the ECG test : 0x55 0xAA 0x04 0x01 0x01 0xF9

Stop the ECG test: 0x55 0xAA 0x04 0x01 0x00 0xFA

Start the NIBP test: 0x55 0xAA 0x04 0x02 0x01 0xF8

Stop the NIBP test: 0x55 0xAA 0x04 0x02 0x00 0xF9

Start the Spo2 test: 0x55 0xAA 0x04 0x03 0x01 0xF7

Stop the Spo2 test: 0x55 0xAA 0x04 0x03 0x00 0xF8

Start the Temperature test: 0x55 0xAA 0x04 0x04 0x01 0xF6

Stop the Temperature test: 0x55 0xAA 0x04 0x04 0x00 0xF7

Set the ECG waves Gain: 0x55 0xAA 0x04 0x07 0x03 0xF1 (ECG gain: 0x03

x1Gain)

Set the ECG mode: 0x55 0xAA 0x04 0x08 0x02 0xF1 (0x02 monitor mode)

Set the NIBP test mode: 0x55 0xAA 0x04 0x09 0x01 0xF1 (: 0x01 Adult mode)

Set the NIBP target pressure valve: 0x55 0xAA 0x04 0x0A 0x4B 0xA6 (the target

pressure = 75 (0x4B) x 2 = 150 mmHg)

Set the Respiration Gain: 0x55 0xAA 0x04 0x0F 0x03 0xE9 (0x03 x1Gain)

Read the version of the software: 0x55 0xAA 0x04 0xFC 0x00 0xFF

Read the version of the hardware: 0x55 0xAA 0x04 0xFC 0x01 0xFE

Start the Spo2 waves output: 0x55 0xAA 0x04 0xFE 0x01 0xFC

Stop the Spo2 waves output: 0x55 0xAA 0x04 0xFE 0x00 0xFD

Start the Respiration waves output: 0x55 0xAA 0x04 0xFF 0x01 0xFB

Stop the Respiration waves output: 0x55 0xAA 0x04 0xFF 0x00 0xFC

2、Module→PC (data format)

paramete	A1	A2	A3	A4	A5	A6	A	A	The transmit
ECG	0x	I	II	III	aVR	aVL	a	V	250
ECG data	0x	ECG	HR	Resp	ST	HRV			1
NIBP data	0x	NIBP	Cuff	SBP	MBP	DBP			2
Spo2 data	0x	Spo2 status	Spo2	Pulse					1
Temperat	0x	Temp	Integer of T1	Decim al of T1	Intege r of T2	Decim al of T2			1
Version	0x	ASCII	ASCII	ASCII	...	ASCII	...	...	N/A
Version	0x	ASCII	ASCII 2	ASCII	...	ASCII	...	...	N/A
Spo2	0x	waves							40
Respirati	0x	waves							50

detailed description:

ECG waves (0x01):

I、II。。。V means Lead I、Lead II。。。Lead V, Range from 0~250, The transmit frequency 250bps

ECG data (0x02):

ECG status (single byte): BIT0~BIT7from low to high

BIT0: ECG signal, 1 means weak; 0 means normal

BIT1: ECG leads , 1means Lead off ,0 means normal

BIT3~BIT2: ECG waves Gain , 00: x0.25 Gain; 01: x0.5Gain; 10: x1Gain;

11: x2Gain;

BIT5~BIT4: ECG mode, 00: operation Mode; 01: monitor Mode; 10: diagnosis

Mode;

BIT7~BIT6: not use;

HR (Heart Rate): From 0~250

RR (Respiration Rate): From 0~250

ST: (signed char), From -100~+100 means -1mV~+1mV, for example: -75means -0.75mV, +55means +0.55mV

HRV code (single byte): (option function)

0x00	Under analysis
0x01	Normal
0x02	Arrest
0x03	Ventricular fibrillation
0x04	R ON T
0x05	Multi Ventricular premature
0x06	Double Ventricular premature
0x07	Ventricular premature
0x08	bigeminy
0x09	trigeminy
0x0A	sychnosphygmia
0x0B	bradycardia
0x0C	missed beat

NIBP data (0x03):

NIBP status (single byte):

BIT1~BIT0: NIBP test mode, 00: Adult mode; 01: Child mode; 10: neonate mode;

BIT5~BIT2: test status;

0000	finished
0001	testing
0010	stoped
0011	the pressure high than the 300mmgh
0100	the cuff is too loose
0101	the time is too long
0110	Error
0111	disturb
1000	can't get the result
1001	]initialization
1010	]initialization finished

BIT7~BIT6: not use;

Pressure value (single byte): Real pressure = pressure value x 2, mmHg

systolic pressure (single byte): Range 0~250, (mmHg)

meanarterypressure (single byte): Range 0~250, (mmHg)

diastolic pressure (single byte): Rang 0~250, (mmHg)

Spo2 data (0x04):

Spo2 status (single byte):

0x00	normal
0x01	sensor off
0x02	no finger
0x03	searching pulse
0x04	searching time is too long

Spo2 (single byte): Range 0~100, (127 means data error)

Pulse Rate (single byte): Range 0~250, (255 means data error)

Temperature Data (0x05):

Temperature (single byte): 0x00 means normal; 0x01 means T1 sensor off; 0x02 means T2 sensor off; 0x03 means T1 and T2 sensor off

Integer of Temp (single byte): from 0~45

Decimal of Temp (single byte): from 0~9

Real Temperature = Integer + (Decimal of Temp / 10),

Spo2 waves (0xFE):

Spo2 waves (single byte): from 0~100

Respiration waves (0xFF):

Respiration waves (single byte): from 0~250

Examples(Pm6750 to PC)

The ECG waves : 0x55 0xAA 0x0A 0x01 0x80 0x80 0x80 0x80 0x80 0x80 0x80 0x74

The ECG parameters: 0x55 0xAA 0x08 0x02 0x0C 0x3C 0x14 0x00 0x01 0x98

The NIBP data: 0x55 0xAA 0x08 0x03 0x00 0x00 0x00 0x00 0x00 0xF4

The Spo2 data : 0x55 0xAA 0x06 0x04 0x01 0x00 0x00 0xF4

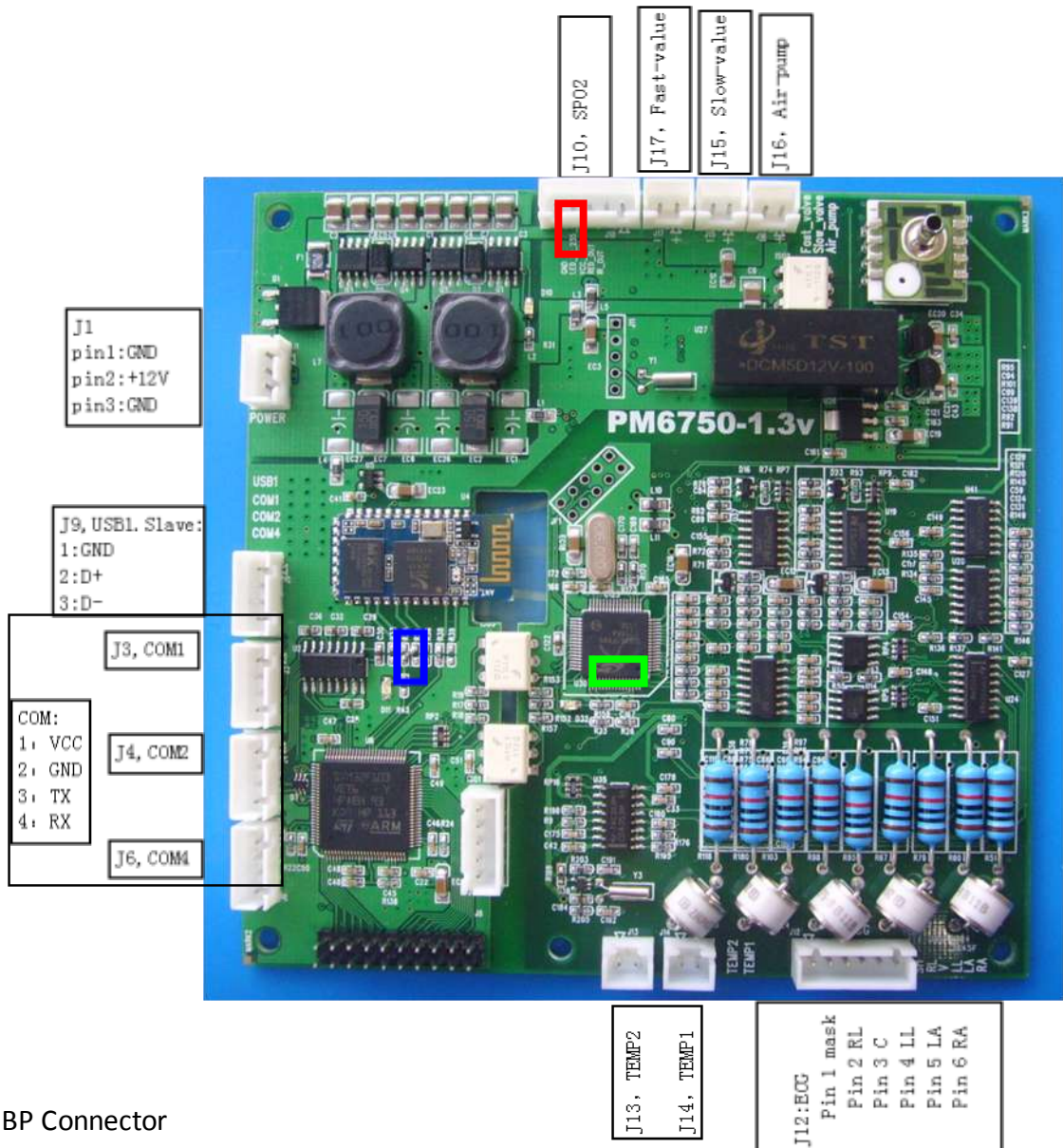
The Temperature data: 0x55 0xAA 0x06 0x05 0x01 0x00 0x00 0xF3

Pm6750 send the version of software: 0x55 0xAA 0x07 0xFC 0x56 0x33 0x2E 0x32 0x13

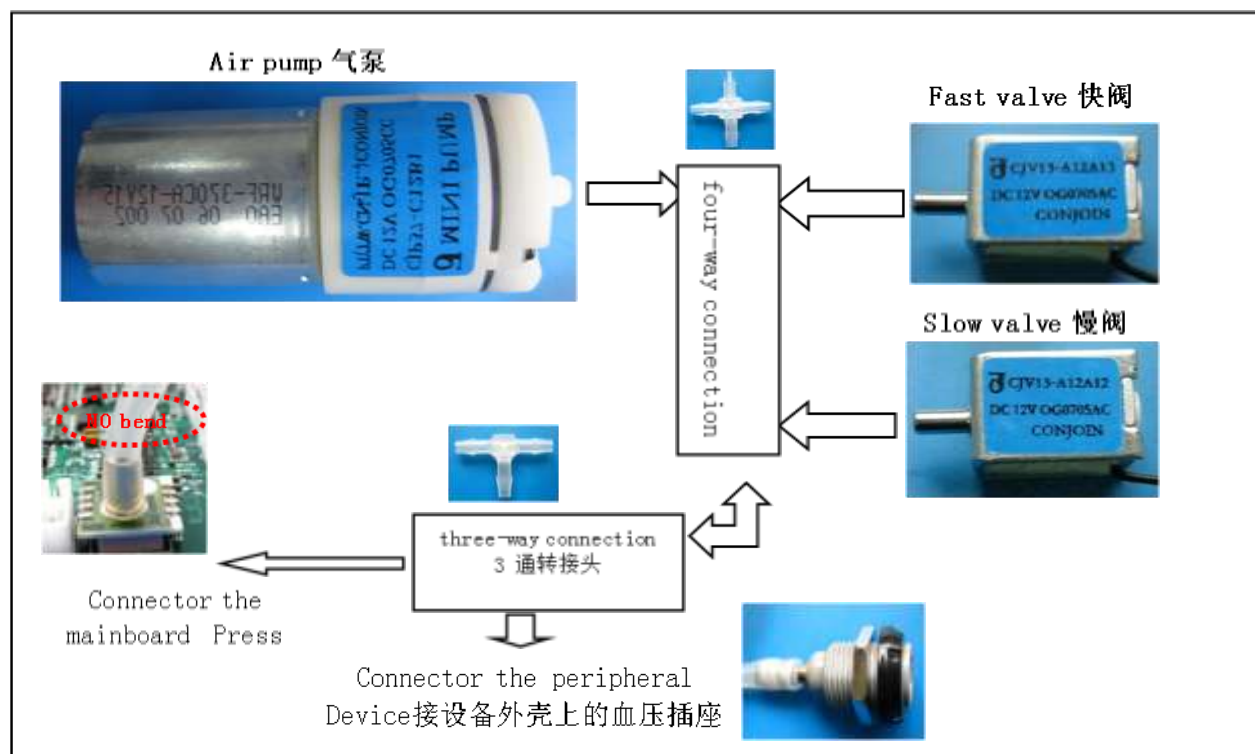
The Spo2 waves : 0x55 0xAA 0x04 0xFE 0x00 0xFD

The Respiration waves : 0x55 0xAA 0x04 0xFF 0x7E 0x7E

4: Connector



5 NIBP Connector



6 Demo software

- 1) Turn on the power for PM6750
- 2) If you use RS232 com, please sure the connect is ok.
- 3) If you use the Bluetooth communication, the Bluetooth is IVT and the PIN code is: 0000
- 4) Set the serial port



START/开始测量



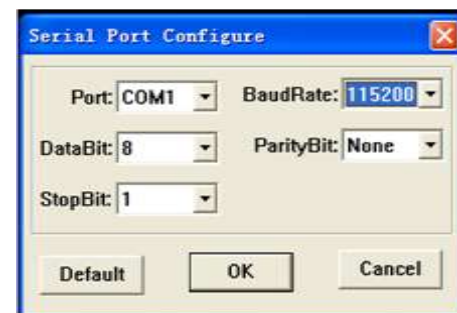
STOP/停止测量



START Blood Pressure Measurement
开始血压测量



START Blood Pressure Measurement
停止血压测量





7 accessories

- 1) Spo2 sensor(soft tip or finger)
- 2) Temp Sensor(Ysi 400 series)
- 3) Nibp cuff and tube
- 4) ECG Leads and cable